MILITARY SPECIFICATION

BEARING, SPIN

(SIDEWINDER)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the spin bearing, a critical item, referred to herein as the "spin bearing" and is part of the guidance and control section, WGU-4A/B (AIM-9M).

1.2 Classification. The spin bearings consist of the following types:

Type I: Cartridge spin bearing.

Type II: Duplex spin-bearing pair.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 3110

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SPECIFICATIONS

FEDERAL

TT-T-656	Tricresyl Phosphate.
QQ-S-763	Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.

MILITARY

MIL-B-197	Bearings, Antifriction; Associated Parts and Subassemblies; Preparation for Delivery of.
MIL-B-1083	Balls, Bearing, Ferrous and Non Ferrous (For use in Bearings and Valves).
MIL-I-6868	Inspection Process, Magnetic Particle.
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice, Process for).
MIL-B-81793	Bearing, Ball, Precision, for Instruments and Rotating Components.
MIL-S-85104	AIM-9M Seeker Assembly (Sidewinder).
MIL-G-85127	Guidance and Control Sections, WGU-4/B and WGU-4A/B Guided Missile (Sidwinder).

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.

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MIL-STD-206	Friction Torque Testing for Instrument Ball Bearings Test Requirements.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1569600	Protector, Dome.
2412395	Sheet, Technical Frequency Specifications.
SA493245	Spin Bearing Preload Fixture.
SA2524273	Slow Down and Noise Test Fixture, Assembly and Details.
639AS3169	Gimbal Assembly.
639AS3180	Gyro Assembly.
639AS3962	Seeker Section, Infrared, Guided Missile.
639AS6502	Gyro Assembly.
639AS6560	Retainer, Ball.
639AS6561	Bearing, Spin.
639AS6562	Polyimide, Porous Molded.
639AS6563	Lubricant, Synthetic Hydrocarbon.

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PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-874	Clean Room Conditions.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ANTI-FRICTION BEARING MANUFACTURERS ASSOCIATION (AFBMA)

AFBMA Standards	Annular Bearings Engineers Committee of the Anti-Friction Bearing Manufacturers Association (Twelve Sections).

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers Association, 60 East 49th St., New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1-1978	Surface Texture.
ANSI Y14.5-1973	Dimensioning and Tolerancing.

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E45	Determining the Inclusion Content of Steel.
ASTM E92	Vickers Hardness of Metallic Materials.
ASTM E384	Microhardness of Materials.

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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS

Handbook H4	Federal Supply Code for Manufacturers, United States and Canada.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Item description. The spin bearing is an inner race duplex preloaded instrument precision annular ball bearing. The spin bearing is of the following two types:

Type I: Cartridge spin bearing. The cartridge spin bearing consists of the following:

a. Double raceway grooves in flanged single outer bearing ring.

b. Two inner bearing rings with a single raceway groove in each ring.

c. Two ball retainers in each cartridge bearing.

d. Twelve balls in each raceway groove for a total of twenty-four balls in each cartridge bearing.

e. Open inseparable assembly.

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Type II: Duplex spin-bearing pair. The duplex spin bearing pair consists of the following:

a. One plain (unflanged), single raceway groove, outer bearing ring.

b. One flanged, single raceway groove, outer bearing ring.

c. Two inner bearing rings with a single raceway groove in each ring.

d. Two ball retainers in each duplex spin bearing.

e. Twelve balls in each individual spin bearing for a total of twenty-four balls in each duplex spin bearing.

f. Individual bearing assemblies are open and inseparable.

3.2 Characteristics.

3.2.1 Performance. The spin bearing shall meet the performance verification characteristics specified herein.

3.2.1.1 Low speed torque (Type I or Type II).

3.2.1.1.1 Non-preloaded spin bearing. The individual inner rings or individual bearings, when operated as single bearings in conjunction with their respective outer ring raceway groove and lubricated with oil only in accordance with 3.3.2.2.3.2 shall display the following torque characteristics (see 6.4):

a.	Average running torque	≤3,000 milligram-millimeter (mg-mm).
b.	Average hash width	≤3,000 mg-mm.
c.	Maximum hash (bearing (contamination)	≤5,500 mg-mm.
d.	Geometric wander	≤1,600 mg-mm.
e.	Brinell damage (metal damage)	No repeatable negative spikes 800 mg-mm greater than the average hash.

f. A nonrecurring positive excursion due to retainer hang-up is acceptable provided the duration is less than one-third revolution.

This requirement shall be verified at the completion of all other testing.

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3.2.1.1.2 Relubrication for low speed dynamometer (LSD). In cases where the oil has been removed from the spin bearing, and the sole purpose is to perform an LSD test to determine its condition after environmental testing, each spin bearing shall be lubricated either by impregnation in accordance with 3.3.2.2.3.2 or by adding two drops of oil, Drawing 639AS6563, prior to testing.

3.2.1.2 Running torque slowdown. After vacuum impregnation in accordance with 3.3.2.2.3.2 and prior to performance of spin bearing slowdown, the bearing shall set at room ambient (18 to 30°C) for not less than 24 hours. After the 24 hour period, the running torque of the spin bearing shall cause the rotor speed to decay from 7,200 to 3,600 revolutions per minute (rpm) in not less than 30 seconds.

3.2.2 Physical characteristics.

3.2.2.1 Balls. The balls shall have a nominal diameter of 0.062500 inch and shall conform to the ball quality requirements of MIL-B-81793, Grade 3.

3.2.2.1.1 Ball complement. The cartridge and duplex bearing shall each have a complement of 24 balls. All balls for each individual bearing within the pair shall be taken from the same unit container.

3.2.2.2 Ball retainer The same ball retainer (see Drawing 639AS6560) shall be used for both raceways in the cartridge or duplex bearing. The ball retainer shall be inner-land piloted (see Figure 1).

3.2.2.2.1 Ball retainer clearance (total running clearance).

a.	Ball retainer to outer raceway land	0.0100 inch nominal.
b.	Ball retainer to inner raceway land	0.0025 to 0.0057 inch.
c.	Ball retainer to balls	0.0020 to 0.0060 inch.

3.2.2.3 Rings. The rings shall conform to Drawing 639AS6561 and AFBMA standards (Section 12), ABEC-9P.

3.2.2.3.1 Wall thickness. The minimum wall thickness at the thinnest cross-section between the raceway and the adjacent inside diameter for inner rings and the adjacent outside diameter for outer rings shall be not less than 0.011 inch.

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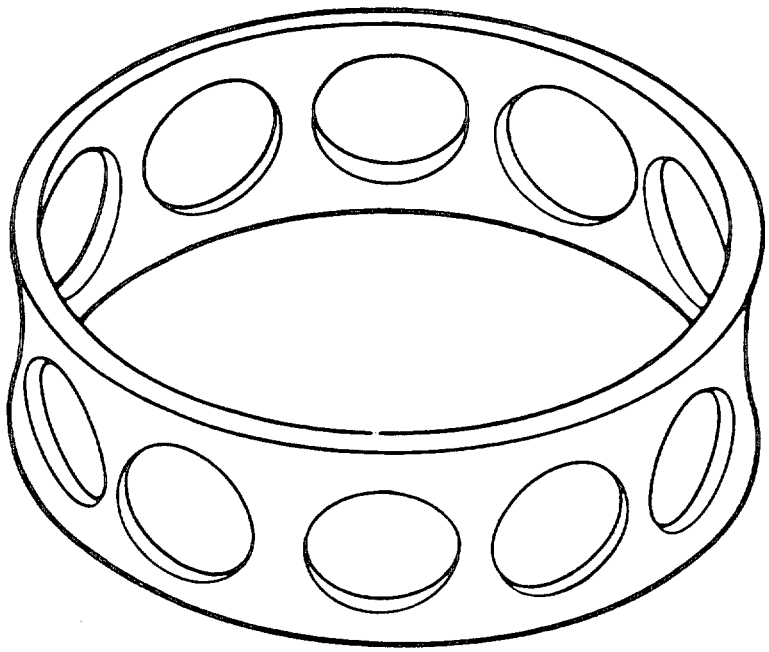


FIGURE 1. Retainer, ball. (See Drawing 639AS6560.)

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3.2.2.4 Raceway radius. The cross-sectional (cross-race) radius of the raceways (race curvature) shall be 56 ±1 percent (0.035 inch nominal) of the nominal ball diameter (0.062500 inch). The maximum departure from a true circle (DFTC) of the cross-race curvature within the working portion or contact area of the ball raceway shall be not greater than 0.000020 inch.

a. Contact angle for plain bearing is 15 to 30 degrees.

b. Contact angle for flanged bearing is 5 to 25 degrees.

3.2.2.4.1 Racewav groove runout. Raceway groove runout shall be not greater than 0.000050 inch in accordance with ABEC-9P tolerances as follows:

a. Outer rings with respect to 0.4062 inch diameter.

b. Inner rings with respect to 0.2344 inch diameter.

c. Race-to-face for all rings with respect to clamping surface "C" (see Drawing 639AS6561).

d. The cartridge flange face is used in lieu of surface "C" for cartridge bearing.

e. Outer raceway parallelism in the outer ring of the cartridge bearing shall be not greater than 0.000050 inch between the two raceways.

3.2.2.4.2 Raceway roundness. The individual circumferential roundness of the inner ring bearing raceway and outer ring bearing raceway, respectively, shall fall within a 0.000025 inch wide annular tolerance zone in accordance with ANSI Y14.5-1973, that is, a fifty-millionths bearing.

3.2.2.5 Individual bearings.

3.2.2.5.1 Inseparable bearings. Individual bearings or inner rings shall be inseparable and shall not separate when a backload of 2.00 ±0.05 pounds is applied through the inner raceway, against the ball dam, at an angle of 0 degree ±15 minutes to the spin reference axis.

3.2.2.5.2 Relative position of bearing components. When the individual bearing is measured under a 1.25 to 2.00 pound preload, the relative positions of the rings and ball retainer shall conform to Drawing 639AS6561 (see 3.2.2.2). The inner ring mating face plane of a mated pair of spin bearings is not required to be coincident with the outer ring mating face plane.

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3.2.2.5.3 Radial play. Radial play shall be as follows:

a. Flanged bearing shall have 0.0006 to 0.0008 inch radial play.

b. Plain bearing shall have 0.0010 to 0.0012 inch radial play.

3.2.2.6 Spin bearings.

3.2.2.6.1 Coding. Both bearings of the spin bearing, Type I or II, shall meet the requirements of 3.2.2.5.2 and shall have the same code designation. Coding shall be in accordance with Drawing 639AS6561 and AFBMA Standards (Section 12). The size code shall be determined by the maximum outside diameter (OD) and the minimum inside diameter (ID). Type I cartridge is treated as a single code outer with matching code inner rings.

3.2.2.6.2 Preloaded pair. The individual bearings or inner rings shall be selected so that the spin bearing shall have a relative DB preload of 1.25 to 2.00 pounds (see Figure 2) (see 3.2.2.6.2.1.1), when installed in accordance with Figure 2 and Table 1. Preload selection for a given set of bearing components shall be compatible with the running torque requirements of 3.2.1.2.

3.2.2.6.2.1 Preload curve plot. The preload curve plot of load (abscissa) versus deflection (ordinate) shall clearly indicate the relative preload present in the bearing pair, Figure 3, regardless of ring orientation. Curve tolerance regardless of preload shall be in accordance with Figure 3.

3.2.2.6.2.1.1 True preload. True preload (see Figure 3) exists when both inner races of the bearing assembly are clamped together in accordance with Table I.

3.2.3 Environmental.

3.2.3.1 Vibration. Upon completion of the vibration test the spin bearing shall be tested in accordance with 3.2.1. Test results shall be not greater than specified requirements by 5 percent.

3.2.3.2 Shock. Upon the completion of the shock test (both operating and nonoperating) the spin bearing shall be tested in accordance with 3.2.1. Test results shall be not greater than specified requirements by 5 percent.

3.2.3.3 Mean-time-between-failure (MTBF) demonstration. Unless otherwise specified in the contract or purchase order (see 6.2.1), spin bearings supplied in accordance with this specification shall be capable of demonstrating a specified MTBF (Θo) of 650 hours when tested in accordance with MIL-STD-781 utilizing Test Plan IC.

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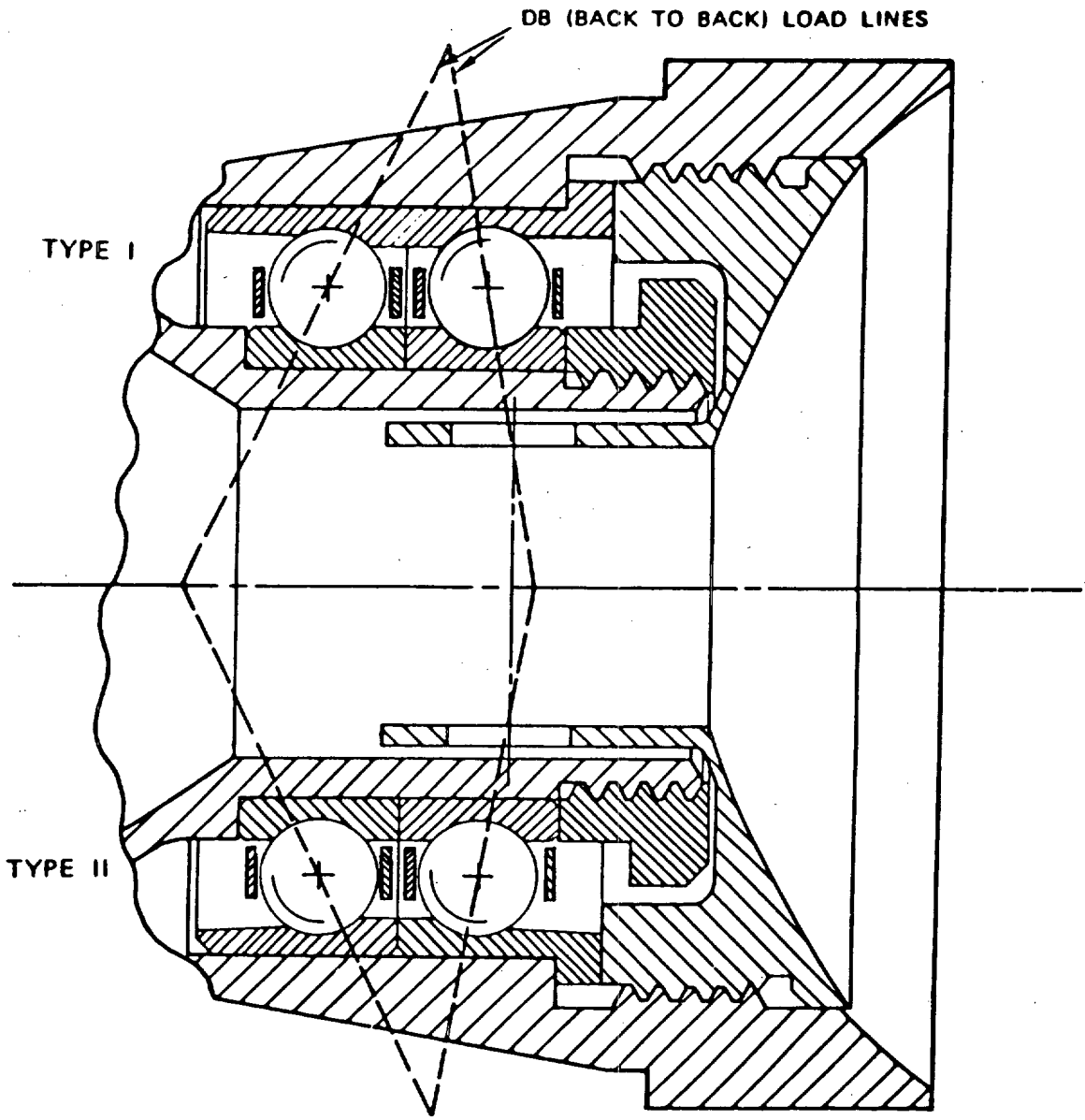


FIGURE 2. DB preload diagram.

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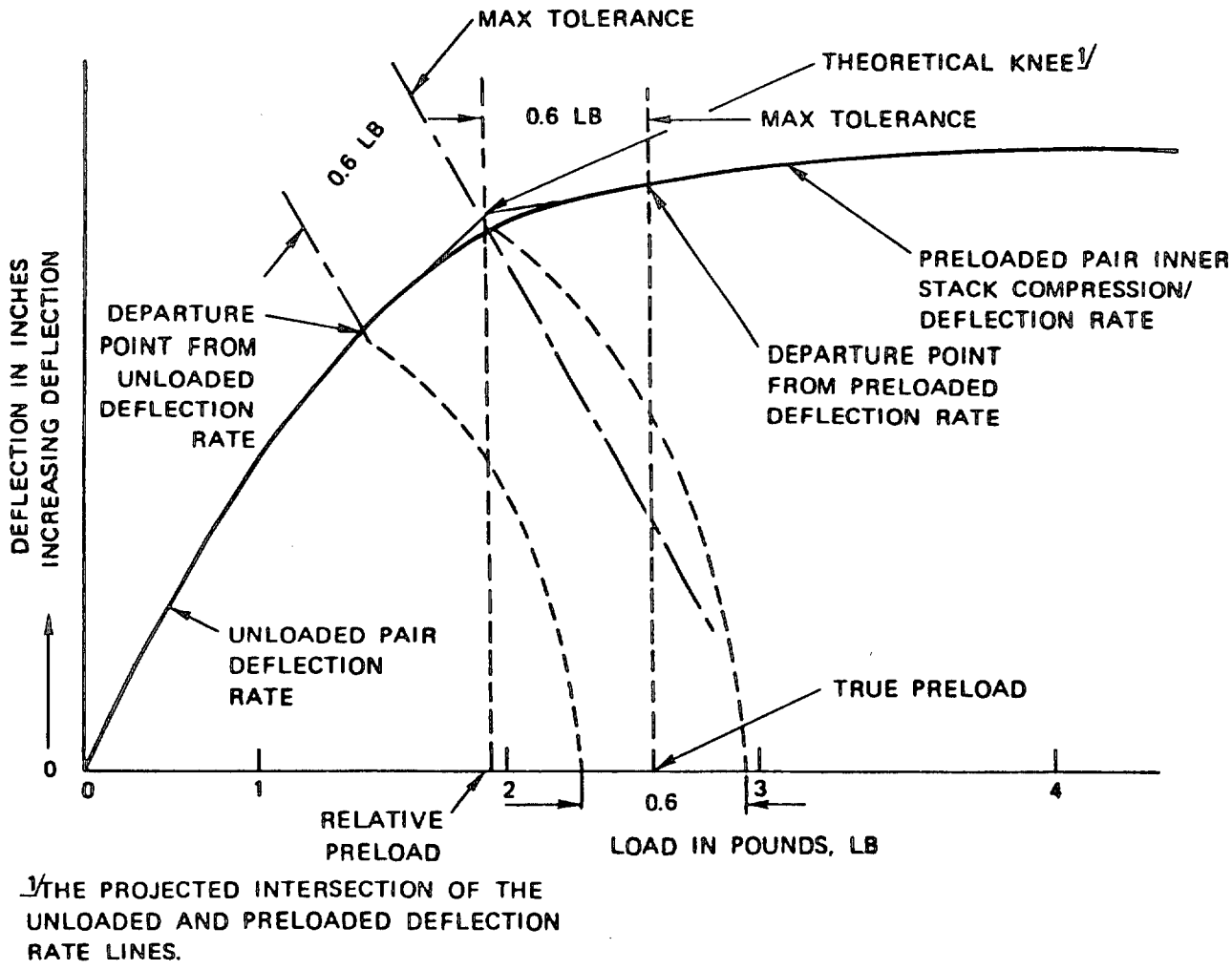


FIGURE 3. Preload curve tolerance.
(Related to Drawing SA493245.)

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TABLE I. Mass properties, fit, and clamping forces.

Gyro or dummy rotor mass (gm)	Gyro or dummy rotor polar moment of inertia ($gm-cm^2$)	Maximum unbalance (micro-ounce-inch)	Bearing shaft clearance (micro-inch)1/	Bearing housing clearance (micro-inch)1/	Inner ring clamping torque (inch-ounces) (based on 0.2344-64) thread)
107 ±2	384 ±3	150	30-80	50-150	30-40

1/ Bearing clearances are determined by using the minimum ID of the hole and the maximum OD of its mating part.

3.2.3.3.1 MTBF performance requirements. The MTBF performance, at the guidance control section (GCS) level during MTBF temperature cycling, shall be as following:

a. Gyro regulated speed - F_1 +3, -4 revolutions per second (rps) (see Drawing 2412395)

b. B+ current - shall be not greater than 1.1 amperes

c. B- voltage - -25 ±3 volts direct current (Vdc)

At the conclusion of MTBF demonstration testing, the spin bearings shall be removed from the GCS and the spin bearings shall be subjected to not less than 12-hour Soxhlet cleaning operation. High purity hexane shall be utilized in the cleaning process. Upon removal of the bearings from the cleaning operation, the bearings shall be subjected to an ambient vacuum drying process for not less than 30 minutes. Lubricate each bearing in accordance with 3.2.1.1.2 and perform testing in accordance with 3.2.1. Test results shall be not greater than specified requirements by more than 10 percent.

3.2.3.3.2 MTBF determination. The spin bearing running time, while operating in a GCS, shall be used in MTBF determination. Each spin bearing running time shall be not less than one-half of the average running time of all spin bearings which were utilized in the MTBF demonstration. For the purpose of MTBF determination, the spin bearing running time shall be that time during the thermal cycle when the GCS is energized.

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3.3 Design and construction.

3.3.1 Production drawings. The spin bearing shall be manufactured, fabricated, and assembled in accordance with the documents specified in Drawing 639AS6561 and as specified herein. Details of the parts not specified by this specification and Drawing 639AS6561 shall be optional with the contractor.

3.3.2 Standards of manufacture.

3.3.2.1 Materials. Materials meeting specified requirements shall be used in the manufacture of component parts and fabrication of bearings furnished in accordance with this specification.

3.3.2.1.1 Reconditioned component parts. Spin bearings purchased as new items in accordance with this specification shall not contain component parts that have previously been installed in any equipment and operated.

3.3.2.1.2 Porosity control. Material for fabrication of the ball retainer shall be in accordance with Drawing 639AS6562.

3.3.2.1.3 Metals. The inner and outer rings shall be made from consumable-electrode, vacuum-melted stainless steel. All steel used in the rings and balls shall conform to QQ-S-763, Class 440 C, Condition A.

3.3.2.1.4 Polyimide. The ball retainer shall be of polyimide rod (Drawing 639AS6562) and manufactured in accordance with Drawing 639AS6560.

3.3.2.2 Processes.

3.3.2.2.1 Heat treatment. Heat treatment shall be in accordance with MIL-H-6875.

3.3.2.2.2 Passivation. The rings and balls shall be passivated, after the last material removal operation, in accordance with MIL-B-81793.

3.3.2.2.2.1 Surface treatment. After passivation and prior to assembly, the rings and balls shall be treated with Tricresyl Phosphate (TCP) $((CH_3D_6H_4O)_3PO)$ in accordance with TT-T-656 to ensure uniform distribution of lubricant on the balls and within the raceways (no oil beading).

3.3.2.2.3 Lubrication.

3.3.2.2.3.1 Oil lubricant. The oil lubricant shall be in accordance with Drawing 639AS6563.

3.3.2.2.3.1.1 Contamination. The oil lubricant contamination shall meet the requirements of Drawing 639AS6563.

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3.3.2.2.3.2 Ball retainer impregnation. Prior to oil application, the bearings shall be thoroughly cleaned. Using a suitable solvent, wash bearings and then subject bearings to a vacuum bake of 65 to 82 degrees Celsius (°C) for not less than 30 minutes. Immediately after vacuum bake, immerse bearings in lubricating oil specified on Drawing 639AS6563, elevate temperature to 65 to 82°C, and evacuate to not greater than 26 inches of mercury for not less than 30 minutes. Allow bearings to return to ambient (18 to 30°C). Remove excess oil from Type II bearings by centrifuging each bearing at 2500 ±50 gravity units (g) with spin axis parallel to direction of force for 5 to 10 minutes. Retained oil after the centrifuge operation on each bearing (flanged or plain) shall be from 3 to 5 milligrams (mg). The cartridge bearing (Type I) shall be lubricated as an individual bearing. Excess oil shall be removed by centrifuging the bearing 2500 g for a 3- to 6-minute period each side, that is, the bearing shall be subjected to the 2500 g for a 3- to 6-minute period on one side and then turned 180 degrees and subjected to a 3- to 6-minute period of 2500 g. Direction of force during the centrifuge operation shall be parallel to the spin axis. Retained oil after centrifuging shall be 6 to 10 mg.

3.3.2.2.4 Solvent purity. Only high purity reagent grade solvents with a maximum residue of five parts per million (ppm) shall be used in all processes involving finished bearings or finished components.

3.3.2.3 Parts.

3.3.2.3.1 Hardness. When heat treated as specified in 3.3.2.2.1, the rings shall be through-hardened to not less than hardness of Rockwell C58. The balls shall be through-hardened to not less than hardness of Rockwell C59.

3.3.2.3.1.1 Fracture grain size. Fracture grain size shall be 7.5 or finer.

3.3.2.3.1.2 Cracks and fractures. Ring and ball integrity shall be in accordance with MIL-B-81793.

3.3.2.3.1.3 Dimensional stability. Dimensional stability shall be in accordance with MIL-B-81793.

3.3.2.4 Surface finish of rings and balls. Careful examination shall reveal no evidence of rust, tool marks on the mounting surfaces, scratches across the races, residual embedded abrasives, or Brinell marks on the load-bearing surfaces (see 3.3.2.6).

3.3.2.5 Surface finish of ball retainers. Parts shall conform to Drawing 639AS6560.

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3.3.2.6 Surface roughness of rings. The surface roughness of the rings shall be not greater than the following:

a.	Raceway surfaces:	The average $\overline{X}$ finish of any manufacturing lot shall be not greater than 1.1 microinches arithmetical average (AA) with no single surface greater than 1.5 microinches AA. 1/
b.	Mounting surfaces: Outside and inside diameters plus clamping and mounting faces (see Figure 2):	8 microinches AA.
c.	Ball retainer pilot lands:	8 microinches AA.
d.	Nonfunctional internal surfaces:	32 microinches AA.
e.	Corners, chambers, fillets, and flange OD:	63 microinches AA.

1/ Cross-race lapping may be employed to achieve this finish.

3.3.2.7 Visual and mechanical. The visual requirements of the spin-bearing characteristics shall be in accordance with MIL-B-81793, except as specified herein.

3.3.2.8 Reliability. Spin bearing reliability shall be in accordance with 3.2.3.3.

3.3.2.9 Identification and marking. The spin-bearing package shall be permanently and legibly identified and marked in accordance with MIL-STD-130 (see 5.3.1).

3.3.2.9.1 Ring identification. Outer faces of inner rings or flanged bearing shall be marked with the manufacturer's identification by electroetch, acid-etch, or soft stamp in accordance with the marking for identification of MIL-B-81793, except as noted herein.

3.3.2.9.2.1 Post marking. Identification marking shall not degrade functional surfaces nor impair functional integrity.

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3.3.3 Material cleanliness. Material cleanliness shall be in accordance with MIL-B-81793, except inclusion levels shall be as indicated below:

Type	JERNKONTORET rating
a. Sulfide:	Thin series 1.5; heavy series 1.0
b. Alumina:	Thin series 1.0; heavy series 1.0
c. Silica or silicate:	Thin series 1.0; heavy series 1.0
d. Globular oxides:	Thin series 1.5; heavy series 1.0

3.3.3.1 Component cleanliness. All tools and fixtures that are used to handle the bearings and all bearing components shall not introduce any contamination.

3.3.4 Special working conditions. The facilities used in the manufacture, testing, and inspection of the spin bearing shall be in accordance with AD-874.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.1.2.1 and 6.2.3).

3.5 Workmanship. Workmanship, finishing, cleaning, and lubrication of spin bearing pairs shall be performed in a manner that will assure compliance with all requirements of this specification and all applicable referenced documents. The contractor shall comply with AD-874 and MIL-B-81793, to assure production of high quality bearings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.1.1 Classification of inspections. The inspection and testing of the bearings shall be in accordance with the following classifications:

a. First article inspection (see 4.1.2.1).

b. Quality conformance inspections (see 4.2)

(1) Inspection lot acceptance criteria (see 4.2.1).

(2) Product verification tests (see 4.2.2).

(3) Component tests (see 4.2.2.1).

4.1.1.2 Test equipment. Test equipment shall be of sufficient accuracy, quality, and quantity to ensure that the performance for the required tests and measurements are to the accuracy specified herein. All equipment shall be calibrated and maintained in accordance with a system meeting the requirements of MIL-STD-45662.

4.1.2 First article inspections.

4.1.2.1 First article inspection sequence. The first article sample (see 3.4) shall be subjected to the following inspection sequence:

a. All 12 spin bearings shall be subjected to all of the acceptance tests listed in Table II, Group I.

b. Six spin bearings shall be selected after completion of the Table II, Group I tests and subjected to all the product verification tests listed in Table II, Group II.

c. Four of the spin bearings used for the Table II, Group II tests shall be further subjected to the environmental tests listed in Table II, Group III.

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d. The remaining two spin bearings used for product verification tests shall be disassembled and the components used for the Table II, Group IV tests. Each of the component tests listed in Table II, Group IV shall be performed on the applicable components of both spin bearings.

e. A total of six spin bearings from the remaining first article pairs shall be subjected to the MTBF demonstration test of Table II, Group V.

f. Failure of the first article sample to pass the tests of Table II shall cause rejection of the sample. In the event of failure during the first article testing and inspections of Table II, Groups I, II, III, or IV, testing shall be stopped and the procuring activity and their technical representatives shall be notified within 24 hours. Complete failure analysis shall be performed and a final report prepared as specified in the contract or purchase order (see 6.2.2). First article testing and inspections may continue only with the concurrence of the procuring activity. In the event of failure during the first article testing of Table II, Group V see paragraph 4.3.3.4.

In the event of a failure during any stage of the first article inspection, failure analysis shall be used to determine if the failure should be attributed to poor manufacturing, design, or to some unrelated cause. Additional inspections shall be performed after failure correction to verify that acceptable bearings are being produced and the failure mode eliminated. The procuring activity must be satisfied that the bearing manufacturer can and does meet the intent and requirements of this specification.

4.2 Quality conformance inspections. The production lot sample shall be subjected to the inspections of Table II and as specified in 4.2.1 and 4.2.2.

4.2.1 Inspection lot acceptance criteria. Each spin bearing shall be subjected to all the tests listed in Table II, Group I, Subgroup 1 in accordance with 4.2.4.1. Any spin bearing which fails any Group I, Subgroup 1 acceptance tests shall be rejected. Each spin bearing of Group I, Subgroup 2 acceptance test sample selected in accordance with 4.2.4.1 shall be subjected to the running torque test of Table II, Group I, Subgroup 2. If any failure occurs, reject the failed spin bearings and subject each spin bearing of the lot represented by the sample to the running torque test of Table II, Group I, Subgroup 2.

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TABLE II. First article and quality conformance inspections.

Inspection group	Test	Requirement paragraph	Method paragraph
Group I			
Acceptance tests			
Subgroup 1	Lubrication	3.3.2.2.3	4.3.4.4.3
	Workmanship	3.3.1, 3.3.2.10, 3.5	4.3.5
	Coding	3.2.2.6.1	4.3.2.1
	Preload	3.2.2.6.2	4.3.2.2
	Low sped torque	3.2.1.1	4.3.1.1.1
Subgroup 2	Running torque	3.2.1.2	4.3.1.2
Group II			
Product verification tests	Inseparable bearings	3.2.2.5.1	4.3.2
	Relative position of bearing bearing components	3.2.2.5.2	4.3.2
	Radial play	3.2.2.5.3	4.3.2
Group III			
Environmental tests	Vibration	3.2.3.1	4.3.3.1
	Shock	3.2.3.2	4.3.3.2
Group IV			
Component-parts tests	Rings, dimensions, and tolerances	3.2.2.3 3.2.2.3.1	4.3.2
	Raceway radius and runout including DFTC for both cross-race and circum-ferential roundness	3.2.2.4, 3.2.2.4.1, 3.2.2.4.2	4.3.2
	Ball, dimensions, and tolerances	3.2.2.1	4.3.2
	Ball complements	3.2.2.1.1	4.3.2
	Ball retainer	3.2.2.2	4.3.2
	Ball retainer porosity	3.3.2.1.2, 3.3.2.2.3.2	
	Surface finish, rings and balls	3.3.2.4	4.3.4.3.2
	Surface finish, retainers	3.3.2.5	4.3.4.3.3
	Surface roughness, rings	3.3.2.6	4.3.4.3.4
	Passivation	3.3.2.2.2	4.3.4.2.1

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TABLE II. First article and quality conformance inspections. (Continued)

Inspection group	Test	Requirement paragraph	Method paragraph
Group IV (Continued)	Dimensional stability	3.3.2.3.1.3	4.3.2 4.3.4.3.1.3
	Hardness	3.3.2.3.1	4.3.4.3.1
	Materials, rings, and balls	3.3.2.1, 3.3.2.1.1, 3.3.2.1.3	4.3.4.1.1
	Materials, ball retainer	3.3.2.1.4	4.3.4.1.2
	Wettability of rings and balls	3.3.2.2.2.1	4.3.4.2.2.1
	Decarburization	3.3.2.4	4.3.4.3.2.1
	Fracture grain size	3.3.2.3.1.1	4.3.4.3.1.1
	Cracks and fractures	3.3.2.3.1.2	4.3.4.3.1.2
	Material cleanliness	3.3.3.1	4.3.5
Group V			
MTBF	MTBF demonstration	3.2.3.3	4.3.3.3

4.2.2 Product verification tests. Each spin bearing of the product verification sample, selected in accordance with 4.2.4.2, shall be subjected to all the tests of Table II, Group II. If any spin bearing fails any Table II, Group II tests, the sample shall be rejected and a new sample shall be drawn and tested. If any failures occur during the testing of the second sample, reject the failed bearing and subject the entire lot represented by the two samples to the test or tests in which the failures occurred.

4.2.2.1 Component tests.

4.2.2.1.1 Prior to assembly. Component-parts selected in accordance with 4.2.4.3 shall be subjected to the applicable tests listed in Table II, Group IV. Acceptance or rejection of the component lot shall be in accordance with 4.2.4.3.

4.2.2.1.2 Test after assembly. Two or more spin bearing pairs from each first article sample shall be dismantled and subjected to the tests listed in Table II, Group IV. Bearings previously used in product verification tests and forwarded to the testing activity in compliance with 4.2.4.2 shall also be subjected to the Table II, Group IV tests, at the discretion

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of the procuring activity. Should any component fail to meet the requirements of the Table II, Group IV tests, the procuring activity reserve the right to retest the bearings or withdraw the qualification of the product, or to take such other remedial action as deemed appropriate.

4.2.3 Lot. A lot shall consist of all spin bearings made of the same materials group and in the same production run which are offered for delivery at the same time. A lot of components shall consist of all like components from the same machine run or production process being submitted for inspection at the same time.

4.2.4 Sampling procedures.

4.2.4.1 Sampling for acceptance tests. Sampling for acceptance tests of Table II, Group I, subgroup 1 shall be on a 100-percent basis. Sampling for acceptance test of Table II, Group I, Subgroup 2 shall be in accordance with MIL-STD-105, Special Inspection Level S-3, Single Sampling Plan, Acceptable Quality Level (AQL) Level 4.0 (see 4.2.1 for defective rate).

4.2.4.2 Sampling for product verification tests. The sample for product verification tests shall be 4 percent of the production bearing lot. On completion of product verification tests, one-half of the test sample shall be held for disposition as specified in the contract or purchase order (see 6.2.1). The remaining half of the samples shall be returned to the lot.

4.2.4.3 Sampling for component-parts tests. Sampling of component parts prior to assembly shall be in accordance with MIL-STD-105, Inspection Level II, Double Sampling Plan. The sample lot shall be accepted or rejected in accordance with AQLs specified in Tables III and IV for the designated category and classification of critical, major, and minor defects specified in MIL-B-81793. Sampling of parts of assembled bearings for product verification or first article shall be as specified in the applicable test requirements.

TABLE III. Categories of defects and AQLs by tolerances.

Tolerance	Defect category	AQL (percent)
ABEC-9P	Critical	0.65
	Major	1.00
	Minor	2.50

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TABLE IV. Classification of defects.

Category	Code	Description	Requirement
	126	Ball retainer material	3.3.2.1.4
	127	Ball diameter variation not within specified limits	3.2.2.1
	128	Balls do not meet visual requirements	3.3.2.4
	129	Ball surface finish exceeds specification	3.2.2.1
	130	Heat treatment does not meet requirements of specification	3.3.2.2.1
Minor	201	Outer ring width variation	3.2.2.3
	202	Outer ring flange width	3.2.2.3
	203	Outer ring flange OD	3.2.2.3
	204	Outer ring corner radii	3.2.2.3
	205	Outer ring surface roughness on mounting surface exceeds specification	3.3.2.4
	206	Outer ring land surface roughness exceeds specification	3.3.2.4
	207	Inner ring width variation	3.2.2.3
	208	Inner ring surface finish on mounting surface exceeds specification	3.3.2.4
	209	Inner ring land surface finish exceeds specification	3.3.2.4
	210	Raceways do not meet visual requirements	3.3.2.7
	211	Ball retainers do not meet visual requirements (other than Major 126)	3.3.2.7
	212	Ball retainer clearance missing	3.2.2.2.1
	213	Mounting surfaces do not meet visual requirements	3.3.2.7
	214	Bearing marking, bearing identification, package marking, or mounting instructions for duplex pairs are incomplete or missing	3.3.2.10 3.3.2.10.1
	215	Bearing packaging or packing not as specified (for coding only refer to MIL-B-197 for inspection details)	5.2, 5.3
	216	Bore and/or OD size coding is incomplete or missing	3.2.2.6.1

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4.3 Test methods.

4.3.1 Performance.

4.3.1.1 Low speed torque. Lubrication for the torque trace test shall be in accordance with 3.3.2.2.3.2. (Data shall be recorded on a chart with a scale sensitivity of 500 mg-mm or less for each division.) The bearing rotation speed and the chart speed combination shall be one of the following conditions:

Condition	A	B	C
Bearing speed (rpm)	1/2	1	2
Chart speed (mm/sec)	50-60	120-150	120-150

4.3.1.1.1 Type I. The individual inner rings, when operated as single bearings, shall be tested in accordance with MIL-STD-206 at a 400-gram (gm) load. The torque trace shall meet the requirements of 3.2.1.1.1.

4.3.1.1.2 Type II. The individual bearings (plain or flanged) of the duplex bearing shall be tested in accordance with MIL-STD-206 at a 400-gm load. The torque trace shall meet the requirements of 3.2.1.1.1.

4.3.1.2 Running torque slowdown. The spin bearing shall be lubricated in accordance with 3.3.2.2.3.2, and installed in a slowdown and noise fixture (see Drawing SA2524273), or equal slowdown test fixture approved by the procuring activity using a dummy rotor having properties, fit, and clamping forces conforming to Table I, except the bearing shaft and housing clearances shall be 100-300 microinches. The running torque of the bearing shall meet the requirements of 3.2.1.2.

4.3.2 Physical. The bearings shall be subjected to the appropriate tests of either AFBMA Standards or MIL-B-81793 and shall meet the physical requirements of 3.2.2.

4.3.2.1 Coding. The bearing shall be measured in accordance with gaging practices specified in AFBMA Standards (Section 12) for instrument ball bearings. The spin bearing shall meet the requirements of 3.2.2.6.1.

4.3.2.2 Preloaded pair. The preload shall be measured on the bearing preload fixture (see Drawing SA493245) or equal preload test fixture approved by the procuring activity with the bearing lubricated as specified in 3.3.2.2.3.2. The preload shall meet the requirements of 3.2.2.6.2.

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4.3.2.2.1 Preload plot. Figure 3 is an illustration of a hypothetical preload plot consisting of two AA straight lines connected by a curve that starts and stops with 0.6 pound on either side of the tangent intersection point (preload point). The bearings shall be rejected when the curved portion of the plot starts and stops outside the limits shown on Figure 3 regardless of preload (see 3.2.2.6.2.1.1).

4.3.3 Environmental. The environmental tests of Table II, Group III, shall be performed on bearings selected from the first article sample. Failure of the sample to meet the requirements specified herein will be cause for further investigation and possible disqualification of the vendor.

4.3.3.1 Vibration. The spin bearing shall be assembled to a gimbal, (see Drawing 639AS3169) which is then installed in a gyro (see Drawing 639AS3180 or 639AS6502) in accordance with the fit, balance, clamping force, and other parameters of Table I, and installed in a seeker assembly (see Drawing 639AS3962), for the vibration test. The seeker assembly shall be mounted on a fixture approved by the cognizant design activity (see 6.2.1 and 6.3). The gyro shall be rotated in accordance with frequency F1 of Drawing 2412395 throughout the vibration test, which will be conducted at an ambient temperature of 18 to 30°C. The seeker assembly shall be subjected to random vibration of 10 minutes duration along each of three mutually perpendicular axes, with inputs as shown on Figure 4. The spin bearing shall meet the requirements of 3.2.3.1 at the conclusion of these tests.

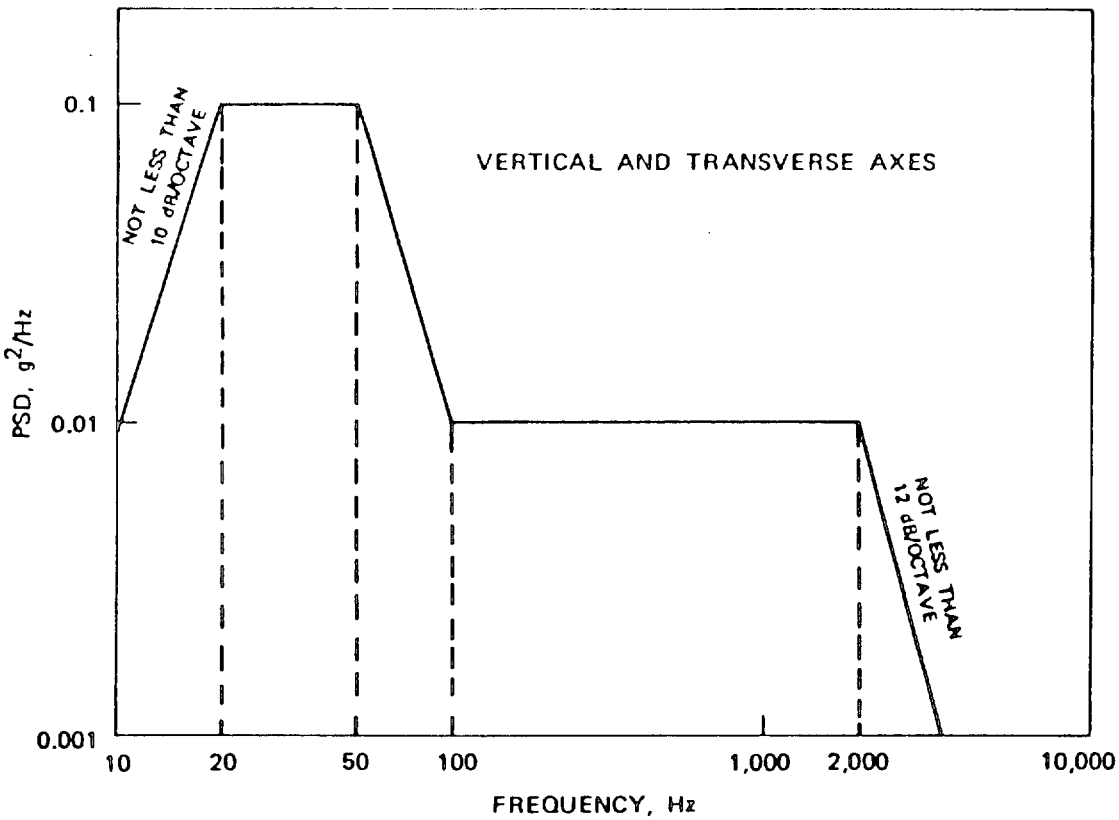
4.3.3.2 Shock. Shock tests shall be divided into two phases, nonoperating and operating.

4.3.3.2.1 Mounting. The spin bearing shall be installed as specified in 4.3.3.1 with adaptability to shock tests along each of the three mutually perpendicular axes. The acceleration of all shocks shall be measured on the aft end of the seeker housing.

4.3.3.2.2 Procedure for shock test (nonoperating). A standard dome protector (see Drawing 1569600), shall be firmly secured to the front end of the seeker head in order to provide magnetic caging on the nonoperating gyro. For this test, the gyro shall possess the magnetic characteristics of a production gyro. Subject the bearing, installed as specified in 4.3.3.1, to 12 shocks, two in each direction, along each axis (four per axis) as specified in Table V and Figure 5.

4.3.3.2.3 Procedure for shock test (operating). Repeat the test specified in 4.3.3.2.2 with the gyro or dummy gyro operating at the spin frequency specified in 4.3.3.1, except the dome protector shall be removed for this test. After completion of the test, the bearing shall meet the requirements of 3.2.3.2.

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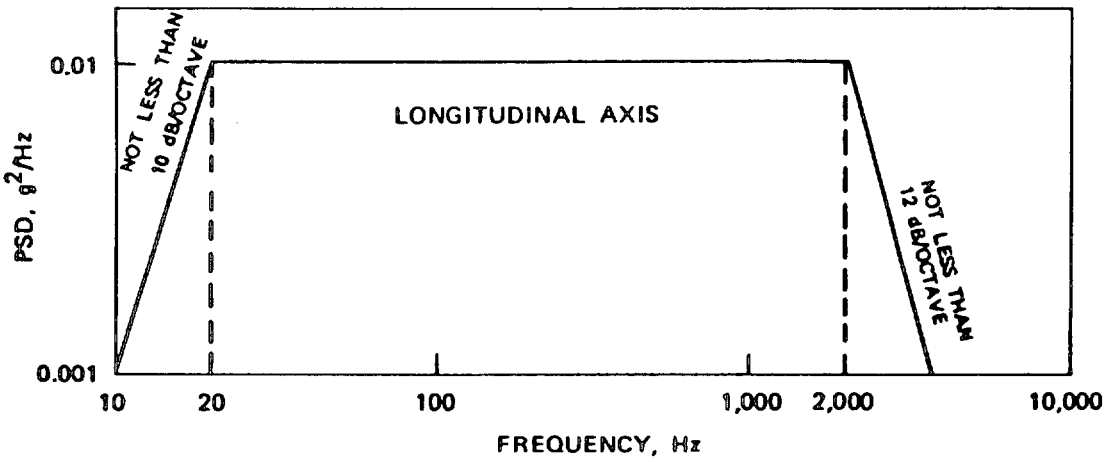


FIGURE 4. Vibration spectrum.

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TABLE V. Shock test.

Direction	Acceleration peak (g) A	Duration of acceleration (milliseconds)	Number of operating shocks	Number of non-operating shocks
Along spin axis	47.5 ±2.5	21.0 ±3.0	4	4
Along outer gimbal axis	47.5 ±2.5	21.0 ±3.0	4	4
Along inner gimbal axis	47.5 ±2.5	21.0 ±3.0	4	4

4.3.3.3 MTBF demonstration. Each spin bearing shall be mounted in a seeker assembly (see Drawing 638AA3962), which shall be subsequently mounted in a GCS which is mounted in a MTBF certified test fixture with the geometric center line of the test fixture located on the vibration exciter center line and subjected to the MTBF demonstration test of 4.3.3.3.1. At the conclusion of the MTBF demonstration, each spin bearing shall meet the requirements of 3.2.3.3.1.

4.3.3.3.1 MTBF demonstration test. The MTBF demonstration test shall be conducted in accordance with MIL-STD-781 utilizing Test Plan IC as modified by 4.3.3.3.2 and the following:

a. Environmental exposures and test profile shall be in accordance with Figure 6. The vibration input shall be in accordance with Figure 6, Note 4.

b. During thermal cycling, the parameters of 3.2.3.3.1 shall be continuously monitored for conformance to requirements.

c. The following performance parameters of MIL-G-85127 and MIL-S-85104 shall be met at the conclusion of not less than one and not greater than six 24-hour test cycles as specified in Figure 6.

(1) GCS current drain - MIL-G-85127

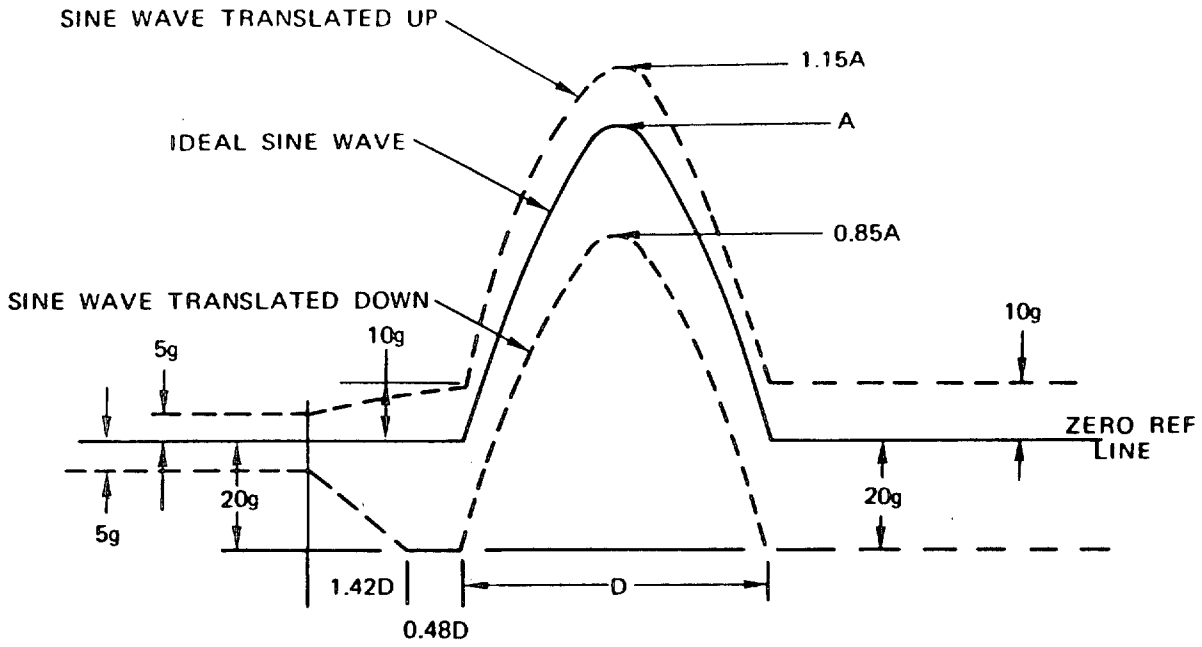
(2) GCS voltage test - MIL-G-85127

(3) Gyro slow down - MIL-S-85104

(4) Gyro speed up (caged) - MIL-G-85127

(5) Gyro regulated speed - MIL-G-85127

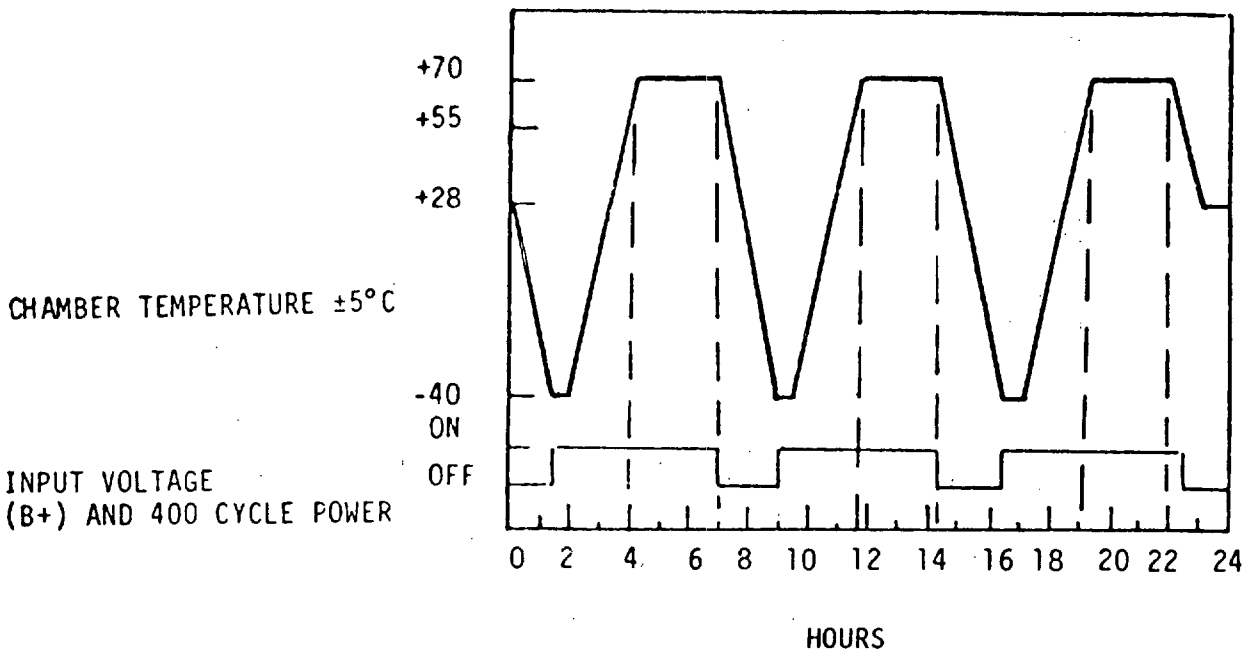
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NOTE:
THE OSCILLOGRAM SHALL INCLUDE A TIME ABOUT 3D LONG WITH A PULSE LOCATED APPROXIMATELY IN THE CENTER. THE ACCELERATION AMPLITUDE OF THE IDEAL HALF SINE PULSE IS A AND ITS DURATION IS D. THE MEASURED ACCELERATION PULSE SHALL BE CONTAINED BETWEEN THE BROKEN LINE BOUNDARIES AND THE MEASURED VELOCITY CHANGE (WHICH MAY BE OBTAINED BY INTEGRATION OF THE ACCELERATION PULSE) SHALL BE WITHIN THE LIMITS $V_i \pm 0.1V_i$ WHERE V_i IS THE VELOCITY-CHANGE ASSOCIATED WITH THE IDEAL PULSE WHICH EQUALS $2AD/\pi$. THE INTEGRATION TO DETERMINE VELOCITY CHANGE SHALL EXTEND FROM 0.4D BEFORE THE PULSE TO 0.1D AFTER THE PULSE.

FIGURE 5. Shock test.

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NOTES:

1. A THERMAL SURVEY MAY BE CONDUCTED IN ACCORDANCE WITH MIL-STD-781 ON ONE UNIT AND NEED NOT BE REPEATED AGAIN. TEMPERATURE STABILIZATION FOR THE UPPER TEMPERATURE (+70°C) ONLY SHALL BE IN ACCORDANCE WITH THE RESULTS OF THE THERMAL SURVEY AND SHALL APPLY TO EACH EXCURSION TO THE PROFILE UPPER TEMPERATURE.

2. INPUT VOLTAGES WILL BE IN ACCORDANCE WITH MIL-G-85127.

3. INPUT VIBRATION LEVEL MONITORED ON VIBRATION FIXTURE.

4. THE VIBRATION PROFILE FOR THE TEMPERATURE CYCLE SHALL BE APPLIED DURING EACH HOUR OF TEST IN WHICH THE GCS IS ENERGIZED. THE PROFILE SHALL BE AS FOLLOWS:

VIBRATION DURATION (MINUTES)	INPUT PSD LEVEL (g^2/Hz)	FREQUENCY (Hz)
50	0.005	20 TO 2000
3	0.015	20 TO 2000
1	0.05	20 TO 100

FIGURE 6. MTBF test cycle.

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(6) Gyro regulated speed (tracking) - MIL-G-85127

(7) Seeker sensitivity test - MIL-G-85127

(8) TRMS dark cell noise test - MIL-G-85127

(9) TRMS noise test (operating) - MIL-G-85127

d. Failure analysis and failure categories shall be as specified in 4.3.3.4.

e. Eighteen hours of GCS operating time shall be accumulated during each 24-hour cycle presented on Figure 6. In the event a failure is detected during chamber testing, all operating time accumulated to the point of failure will be counted. In the event a failure is detected during performance tests, the failure will be assumed to have occurred half-way between the point of detection and the last successful performance test. All operating time and all relevant failure of all spin bearings committed to MTBF testing shall be used in MTBF determination. In the event of failure, the failed unit shall be subjected to the failure analysis of 4.3.3.4.

4.3.3.3.2 Acceptance/rejection criteria. Acceptance/rejection criteria shall be in accordance with MIL-STD-781, Test Plan IC except as modified in Table VI.

TABLE VI. Total test time.

Number of failures	Reject (equal or less)	Accept (equal or more)
0	N/A	6.60 (2,860 scorable hours)
1	N/A	7.82 (3,389 scorable hours)
2	0.68 (295 scorable hours)	9.03 (3,913 scorable hours)
3	9.03 (3.913 scorable hours)	N/A

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4.3.3.4 Failure analysis and failure categories. In the event of a MTBF failure that is verified to be gyro/seeker related, the procuring activity and their technical representatives shall be notified within 24 hours of the original failure date. Complete failure analysis shall be performed and a final report prepared as specified in the contract or purchase order (see 6.2.2). MTBF testing shall not be concluded until failure relevancy is concurred with by the procuring activity. Failure categories shall be as follows:

a. A relevant failure is any failure except those classified and listed herein as nonrelevant failures.

b. Nonrelevant failures are as follows:

(1) Any failure not related to gyro/spin bearing performance.

(2) Any failure caused by test equipment.

(3) Any failure that resulted from operational or procedural error in testing.

(4) Any failure directly resulting from rework, repair, or investigation of prior MTBF failures if detected prior to resubmission of MTBF testing.

(5) Secondary failure when they are confirmed to be induced by failure of another part. However, at least one primary (relevant) failure must be charged when one secondary failure is claimed. Simultaneous failures shall not be classified as primary and secondary failures.

4.3.4 Materials, processes, and parts.

4.3.4.1 Materials.

4.3.4.1.1 Metals. The materials used in the rings and balls shall be tested in accordance with MIL-B-1083, except the magnetic particle inspection shall not be performed. The amount and type of microscopic inclusions shall be determined using the ASTM inclusion chart, Plate III, Designation E45. The material shall meet the requirements of 3.3.3. Certified copies of test reports will be accepted as evidence of compliance with these requirements.

4.3.4.1.2 Polyimide. The polyimide material used in the ball retainer shall meet the requirements of 3.3.2.1.4. Certified copies of test reports and other data from the original source of supply will be accepted as evidence of compliance with this requirement.

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4.3.4.2 Processes.

4.3.4.2.1 Passivation. The rings and balls shall be tested in accordance with MIL-B-81793 and shall meet the requirements of 3.3.2.2.2.

4.3.4.2.2 Surface treatment.

4.3.4.2.2.1 Wettability. TCP treated rings and balls shall be individually tested for lubricant wettability. Sampling for inspection of wettability shall be in accordance with MIL-STD-105, Special Inspection Level S-3, Single Sampling Plan, AQL Level 4.0. Rings and balls shall be dipped in an 85:1 solution of freon and oil (see 3.3.2.2.3.2) and baked at 74 ±5°C for 15 to 20 minutes to remove freon solvent. Examine individual raceways and balls for oil beading or retraction. The requirements of 3.3.2.2.2.1 shall be demonstrated.

4.3.4.3 Parts.

4.3.4.3.1 Hardness. The test procedures and conversion tables of ASTM E92 or ASTM E384, as applicable, shall be used in testing the rings and balls for hardness. The Rockwell C hardness shall be obtained by converting from the Diamond Pyramid Hardness (DPH) as measured on the flat surface of a properly prepared metallographical specimen, using a 500-gm load or by an acceptable equal method. Not less than five readings shall be taken on each ring, and the average value converted to the Rockwell C hardness. Not less than five balls from the same bearing shall be measured, one reading being taken near the center of the flat place on each ball, and the second reading to be taken near the edge of the flat. The hardness of the parts shall be in accordance with 3.3.2.3.1.

4.3.4.3.1.1 Fracture grain size. Fracture grain size test shall be in accordance with MIL-B-1083 for composition 2 balls. The balls shall meet the requirements of 3.3.2.3.1.1.

4.3.4.3.1.2 Cracks and fractures. Test for cracks and fractures using magnetic particle inspection shall be in accordance with MIL-I-6868. The following test conditions apply:

a. Circular and longitudinal magnetization to uncover all possible flaws.

b. Continuous method of particle application.

c. Wet, visible particles.

Indications of surface or subsurface cracks, fractures, or seams shall reject the lot in accordance with 3.3.2.3.1.2.

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4.3.4.3.1.3 Dimensional stability. Dimensional stability shall be tested in accordance with MIL-B-81793. The balls shall meet the requirements of 3.3.2.3.1.3.

4.3.4.3.2 Surface roughness of rings and balls. When inspected for roughness-height rating in accordance with ANSI B46.1-1978, using a 0.003-inch cutoff width and a 0.0001-inch radius diamond stylus, the rings and balls shall meet the requirements of 3.3.2.6 and 3.2.2.1, respectively.

4.3.4.3.2.1 Decarburization. A sample of balls and rings shall be selected in accordance with MIL-STD-105, Single Sampling Plan, Inspection Level S-4, AQL 1.5 percent defective. Transverse sections through the center of each sample shall be polished, microetched, and examined at a magnification of 100 power (X). If any of the test specimens exhibits surface decarburization, the lot shall be rejected.

4.3.4.3.3 Surface finish of ball retainers. The clean, dry ball retainers shall be examined under a 30X magnification microscope and the ball retainers shall meet the surface finish requirements of 3.3.2.5.

4.3.4.3.4 Surface finish of rings. The clean, dry ring shall be examined under a 30X magnification microscope and the rings shall meet the requirements of 3.3.2.6 except for 3.3.2.6a.

4.3.4.3.5 Abrasive examination. Inspect raceways and balls with a 100X microscope illuminated with polarized light. Abrasive particles will show up as bright spots in the dark field. Components shall be free of residual embedded abrasives in accordance with 3.3.2.4. Alternative inspection techniques shall be acceptable to the cognizant design activity.

4.3.4.4 Lubrication.

4.3.4.4.1 Contamination. Lubricant contamination shall be determined in accordance with MIL-B-81793 and shall meet the requirements of 3.3.2.2.3.1.1.

4.3.4.4.2 Ball retainer impregnation. The ball retainer shall meet the requirements of 3.3.2.2.3.2. When specified in the contract or purchase order (see 6.2.2), certification that the requirements of 3.3.2.2.3.2 have been met shall be prepared.

4.3.4.4.3 Oil application. When specified in the contract or purchase order (see 6.2.2), certification that the requirements of 3.3.2.2.3.2 have been met shall be prepared.

4.3.5 Visual and mechanical. The bearing shall be examined for conformance to 3.3.1, 3.3.2.10 and 3.5.

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4.3.6 Preservation, packaging, and marking. Visual examination shall be made prior to shipment to determine conformance with Section 5.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. The individual bearings of the spin bearing pair and data card (see 5.3.1), shall be separately packaged in clean polyethylene and heat sealed in accordance with MIL-B-197, Method 1A-8. The packages shall be sealed for shipment in a foil-lined kraft bag.

5.1.1.1 Alternate method. Pairs of spin bearings shall be packaged in plastic vials in accordance with MIL-B-197, Method 1A-13, with suitable buffer material to prevent movement of the spin bearing pairs in the vial. The vial shall be marked with a pressure-sensitive label (see 5.3.1). Each vial shall be sealed in plastic celon which encloses the pressure-sensitive label.

5.2 Packing.

5.2.1 Level C. Spin bearings, packaged as specified in 5.3.1, shall be packed in accordance with MIL-B-197, domestic shipment for immediate use.

5.3 Marking.

5.3.1 Packaging. Each spin bearing pair shall be accompanied by a data card or label. The data card or label shall contain the following (complete with data):

a. Manufacturer's name and symbol or code identification from Federal Cataloging Handbook H4.

b. NAVAIR Code Identification 30003.

c. NAVAIR drawing number.

d. A unique serial number.

e. Preload.

f. Bore/OD codes or dash number.

g. Lubrication.

h. Manufacturing date.

Any other information requested by the contract or purchase order (see 6.2.1).

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5.3.2 Shipping containers. In addition to any special markings required by the contract or purchase order (see 6.2.1), shipping containers shall be marked in accordance with MIL-STD-129, except there is no NSN for this bearing.

6. NOTES

6.1 Intended use. The bearings specified herein are intended for use as a DB-mounted, preloaded, rotor spin bearing in a gyro optical system used on Guidance and Control Section, WGU-4A/B.

6.2 Ordering data.

6.2.1 Acquistion requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. MTBF demonstration, if required (see 3.2.3.3).

c. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

d. Disposition of samples (see 4.2.4.2).

e. Approval of test fixture (see 4.3.3.1).

f. Packaging information, if other than as specified in 5.3.1.

g. Special markings on shipping containers (see 5.3.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.2.1, 4.3.3.4	Failure reports	UDI-R-21141A	---
4.3.4.4.2	Certification	UDI-A-23264B	---
4.3.4.4.3	Certification	UDI-A-23264B	---

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(DIDs related to this specification and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 12 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, should furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Mounting for shock and vibration. Mounting of the bearing is in accordance with MIL-STD-2081 and MIL-STD-2082.

6.4 Definitions.

6.4.1 Torque test. Torque test definitions that are not included in MIL-STD-206.

6.4.1.1 Average hash width. The average bandwidth of the bearing torque observed during test cycle.

6.4.1.2 Bearing contamination. The torque value of the largest single positive spike peak-to-peak observed during test cycle.

6.4.1.3 Metal damage (Brinell). The torque value of the largest negative single peak-to-peak observed during test cycle.

6.4.1.4 Geometric wander. The variation in average torque during a test cycle, and is the difference between the maximum and the minimum instantaneous average torque during a test cycle.

6.4.2 Bearing and/or spin bearing. These terms are used interchangeably.

Preparing activity:
NAVY-AS
(Project 3110-N002)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-85684(AS)	BEARING, SPIN (SIDEWINDER)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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